

COASTS, ISLANDS, AND REEFS

The world’s continents and islands are all surrounded by coastlines. North America has the longest coast of any continent with 193,000 miles (310,000km), and Canada is the individual country with the longest at 130,500 miles (210,000km). Coastal features include gentle shores with a broad intertidal zone, sand

dunes, vertical cliffs, rock stacks and arches, mangrove swamps, salt marshes, and wide estuaries. Some are deeply indented, especially where sea-level rise has drowned coastal valleys. Coasts advance where deposition exceeds erosion, but recede where marine erosion is greater than deposition.

North America

Ilulissat Icefjord

Location Near Ilulissat, west coast of Greenland

This tidal fjord opening into the Davis Strait is the outlet for the Ilulissat Glacier (also known as Sermeq Kujalleq), the fastest-moving glacier outside Antarctica, pushing into the fjord at 65–115 ft (20–35 m) per day. As the glacier has retreated, the fjord’s open water has extended farther inland, and it is now about 30 miles (50km) in length. It cuts through gneiss, granite, and mica schist 2.5–1.6 billion years old. The ice-carved fjord is up to 3,300 ft

(1,000 m) deep, but much shallower at the iceberg bank, where it opens into the ocean; many of the larger icebergs run aground there.

Puget Sound

Location Washington, western U.S.

This huge complex of tidal basins and channels is the US’s second largest estuary, stretching 100 miles (160 km) from north to south. Its waters connect with those of the Strait of Juan de Fuca and the Pacific Ocean at Admiralty Inlet. Puget Sound’s deep basins, which plunge to 920 ft (280 m) below sea level, were carved and scoured by glaciers. Shallower parts of the estuary mark where there are glacial deposits on the seafloor.

Dungeness Spit

Location North coast of Washington state, western U.S.

Stretching 6 miles (9 km) into the Strait of Juan de Fuca, this is the longest natural spit in the U.S. Prevailing westerly winds drive longshore drift, which has extended the sandy spit by an average of 15 ft (4.5 m) per year over the past 120 years. It is vulnerable to the destructive action of the ocean and was temporarily broken into three sections by a storm in 2001.

Mount Desert Island

Location Maine, eastern U.S.

This is the second largest island on the U.S.’s eastern seaboard. Much of it is granite, the remnants of an ancient igneous intrusion. More recently, it has been shaped by the erosive action of glaciers, which carved out Somes Sound, a deep inlet that almost cuts the island in two. This is the only fjordlike inlet on the east coast of the U.S. Most of Mount Desert Island is in Acadia National Park.

Long Island

Location New York, eastern U.S.

The largest island on the east coast of the U.S., Long Island stretches 120 miles (190 km) from

east to west and is separated from the mainland by the East River and Long Island Sound. Long Island is also incredibly diverse, ranging from the New York City boroughs of Brooklyn and Queens to the wilderness of the Pine Barrens temperate coniferous forest. Much of the surface geology is made up of moraines and other glacial deposits.

Lighthouse Reef

Location Belize coast, Caribbean Sea

Oblong in shape, 22 miles (35 km) from north to south, and generally less than 20 ft (6 m) deep, Lighthouse Reef is a large atoll, part of the Belize Barrier Reef. Divers consider it one of the best dive sites on Earth. Near the center is the Great Blue Hole—a large sinkhole that plunges more than 330 ft (100 m) below sea level. It was formed during the last ice age when the sea level was much lower and the surrounding limestone was dry land. The atoll has an amazing variety of corals and fish, including sharks, and sea turtles, which breed here.

Other coasts, islands, and reefs in North America

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